Fiscal Year:	FY 2013	**Task Last Updated:**	FY 01/30/2013
PI Name:	Natarajan, Mohan Ph.D.		
Project Title:	Targeting Protein Cross Talk Signaling as a Biological Predictor for Space Radiation Cancer Risk		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
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Space Biology Special Category:	None		
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Zip Code:	78229-3901	**Congressional District:**	21
Comments:			
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Key Personnel Changes/Previous PI:	None		
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Task Description:	One of the important concerns of 'late effects' after space radiation exposure is the onset of cancers. The onset of cancer could be initiated by radiation-induced molecular changes that are persistently maintained through a positive feed back cycle and amplified over time. We propose that two key proteins, nuclear factor kappa Beta (NF-kB) and tumor necrosis factor alpha (TNF-a), sequentially co-ordinate and control this cascade of events. A novel role of these proteins in radiation-induced tumor development is identified through our recent in vitro and in vivo studies after low LET radiation exposure. In this project we address: (i) how low doses of high LET radiation trigger initial damage signals that remain persistent and are maintained in the memory of the cell/tissue, (ii) how these signals are amplified, (iii) how these amplified signals are transmitted to result in a sustained radial expansion of initial damage, and finally (iv) how radial communication of these signals activates factors that are responsible for predisposing the susceptible cells to carcinogenesis. We will examine these objectives using both in vitro (parallel plate flow model) and ex-vivo (mouse-artery organ culture) approaches, in which irradiated normal vascular endothelial cells in an intact tissue setup or organ culture will be allowed to communicate with the co-existing non-targeted normal or pro-tumorigenic cells from lung, skin, and breast tissue. An in vivo animal model will be used to prove the tumorigenic potentials of transformed cells. We will compare the outcomes with regard to the LETs (17, 51, and 181 keV/um) of three different HZE ion beams (16O, 28Si, and 56Fe) accelerated to the same velocity (600 MeV/amu) and having similar track structure dimensions, but different ionization densities. This project addresses four of the seven major risks and gaps (Cancer 1, 2, 3, & 4) identified in the Space Radiation risks.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We proposed non-targeted effect through the feed-back cycling of two key proteins, nuclear factor kappa B (NF-kB) and tumor necrosis factor alpha (TNF-a) initiates activation of cancer pre-disposing factors, genomic instability and survival advantage. This study will clarify whether radiation-induced molecular changes that are persistently maintained through positive feed-back cycle are amplified over time. Second, whether those amplified signals through the positive feed back cycle are responsible for clonal expansion of cells that are prone to be tumorigenic. In terms of usefulness of the above study on Earth, normal tissue response is one of the major focuses to achieve acceptable therapeutic gain in radiotherapy of many cancers. Identification and characterization of the molecular mediators that play a role in radiation-induced bystander effects that could lead to genomic instability and survival advantage after therapeutic doses of low LET radiation is important to understand the radiation-induced delayed effect in normal tissue. Therefore the findings from the study proposed with space radiation can set a stage to understand the underlying mechanism of tumorigenesis, tumor recurrence and tumor growth at the treatment site.
Task Progress:	In this initial ten months of the funding period of this project we performed experiments proposed in Aim 1. Aim 1: To study the significance of low doses (0, 0.2, 0.4 or 0.8 Gy) of high LET radiation exposure on an altered autocrine mechanism that initiates, amplifies, and sustains a positive feedback cycle, i.e. triggering NF-kB- TNF-a-NF-kB signaling in the targeted cells. We proposed to compare the outcomes with regard to the LETs (17, 51, and 181 keV/mm) of three different HZE ion beams (56Fe, 28Si and 16O) accelerated to the same velocity (600 MeV/amu), and having similar track structure dimensions, but different ionization densities. Primary aortic endothelial cells obtained from Cell ApplicationTM were cultured and sub cultured continually in endothelial growth medium containing 10% growth supplement reagent (Cell Application Inc. CA) until passage 5. Cells at passage 5 were used for all the experiments with 56Fe and 28Si. (i) When the cells grown to 70% confluent they were exposed to 56Fe/600 MeV/amu to a total dose of 0.1, 0.2, 0.4 or 0.8 Gy. Two flasks were exposed at each time. A total of 56 x T-75 flasks were exposed for these experiments. For each dose, cells from duplicate flasks were harvested at 15 min., 3h, 6h, 12, and 24h. Cells only exposed to 0.8 Gy were also harvested for later time points (36, 48 and 72 h). Nuclear extracts were prepared from these samples as described in our earlier publications. The nuclear protein are being analyzed for the activation of nuclear factor kappa B (NF-kB) by electrophoretic mobility shift assay (EMSA). (ii) The supernatants from those harvest described above were carefully collected, spun down to remove any cell debris, and frozen down at -80oC. These supernatants are currently analyzed for the radiation-induced time-dependent secretion of tumor necrosis factor alpha (TNF-a). The kinetics of TNF-a synthesis and secretion upon radiation exposure are being determined by examining the levels of secreted TNF-a into the medium by TNF-a bioactive assay using L929 cells. If the synthesized protein is at an undetectable level, the culture supernatant will be concentrated 5- or 10-fold using Microcon 10 concentrators (Amicon Inc., Beverly, MA). (iii) Duplicate flasks were exposed at 0.8 Gy 56Fe/600 MeV/amu. Twelve hours post exposure, the cells were treated with TNF-a antibody to block TNF-a signaling. The nuclear extracts prepared from the cells are being analyzed to examine the second phase activation of NF-kB after radiation is, in fact, mediated by soluble TNF-a secreted out of the cells through autocrine mechanism. (iv) Cells in duplicate T-75 flasks were first treated with NF-kB inhibitor isohelenin for 1 h prior to exposure to 0.8 Gy 56Fe/600 MeV/amu. The cells were then harvested 24h post-exposure to examine the secreted TNF-a in the supernatant is dependent on NF-kB activation after radiation exposure. The analysis part of those experiments described in step 2 and 3 is currently ongoing, which will determine the positive feedback mechanism between NF-kB and TNF-a. (v) Next, six sets of duplicate flasks (T-75) were treated with kinase, protease and proteosome inhibitors for 30 minutes prior to radiation exposure to block upstream signaling to NF-kB activation. The cells were then exposed to 0.8 Gy 56Fe/600 MeV/amu. The cells were then harvested 3 h post-exposure to analyze AKT-, MEK-, p38-, IKK-2-, and IkB-dependent activation of NF-kB. We anticipate completing these ongoing experiments on expression and kinase activity by the middle of year 2. The results on completion will allow us to understand whether the high LET radiation-induced NF-kB activation is through canonical or non-canonical pathway and differ from low LET radiation or chemical inducers. (vi) Those series of experiments carried out with 56Fe/600 MeV/amu were completely repeated with 28Si/600 MeV/amu to compare the differences or the similarities between two ions accelerated to the same velocity. These samples are currently being analyzed. Second, while waiting for the approval of the beam time and exposure scheduling, the concept proposed was validated

	using low LET radiation (137Cs) at low doses (0.1 -2 Gy). This approach is important since the results can be compared with the high LET radiation data for their similarities and differences. Further, the study with low LET radiation would also allow us to be prepared with all the molecular tools and optimize all the methods proposed for experiments with the high LET radiation exposure. DNA binding activity and the bioactive assays performed indicated that radiation triggers the activation of the key protein NF-kB which in turn induces the expression and secretion of the pro-inflammatory cytokine, TNF-a. Secreted soluble TNF-a then stimulates the second phase NF-kB activation through an autocrine mechanism. These results clearly revealed that recycling of these two proteins through a positive-feedback mechanism sustain the initial radiation signals and maintains a 'memory' of the initial insult. While TNF-a and NF-kB can induce one another through positive feedback mechanisms, secreted TNF-a initiates an identical TNF-a-NF-kB feedback loop in the bystander cells. These results show how the initial signals are amplified, and transmitted to non-targeted cells resulting in a sustained radial expansion. These findings are important to prove the 'proof of principal' and set a stage to determine how these molecular mediators are functionally interrelated and how they coordinately provide a niche for the development of cancer after high LET radiation exposure. The ongoing experiments on the cells exposed to 56Fe and 28Si at the 12C run are anticipated to be completed in the next four months. Third, in order to address the hypothesis that the cells grown in static culture respond differently to radiation exposure compared to cells grown in their micro-environmental niche, we cultured the cells in parallel plate flow system, exposed to radiation and examined their morphology, endothelial specific gene expression and NF-kB signaling pathway array. This would allow us to understand the combined effect of supporting factors such as extracellular matrix and local environment in which the cells/tissue reside; in particular, with endothelial cells, where the cells are constantly undergoing different types of shear stress due to continuous blood flow. More importantly, this approach will allow us to understand how the cells grown in their micro environmental niche respond to high LET radiation and interpret the results more closely to the in vivo situation. Structural and functional characteristics of endothelial cells maintained under simulated micro-environmental conditions were lost in cells grown in static cultures. Similarly the gene expression studies clearly indicated a difference in the gene expression pattern for both endothelial cell markers and NF-kB signaling pathway. These differences significantly influenced the responses of these cells to low LET radiation.
Bibliography Type:	Description: (Last Updated: 04/11/2021)
Abstracts for Journals and Proceedings	Natarajan M, Yu H, Mohan S. "Protein cross-talk signaling regulates radiation-induced bystander effects and initiates cancer pre-disposing determinants." 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. , Jul-2012
Articles in Peer-reviewed Journals	Yu H, Mohan S, Natarajan M. "Radiation-triggered NF-kappa beta activation is responsible for the angiogenic signaling pathway and neovascularization for breast cancer cell proliferation and growth." Breast Cancer (Auckl). 2012;6:125-35. Epub 2012 Jul 23. http://dx.doi.org/10.4137/BCBCR.S9592 ; PubMed PMID: 22872788 , Jul-2012